

Metlapalli Mehereswar

+91 7702595911 | mehereswar1318@gmail.com | [linkedin.com/in/meher-eswar](https://www.linkedin.com/in/meher-eswar) | github.com/mehereswar24

CAREER OBJECTIVE

Motivated college student with a strong interest in Web Development and Artificial Intelligence/Machine Learning. Proficient in Python and JavaScript with foundational knowledge of Data Structures and Algorithms. Seeking an entry-level Software Development Engineer (SDE) role or internship to apply technical skills to real-world projects and grow as a software engineer.

EDUCATION

Vellore Institute of Technology <i>Bachelor of Technology in CSE (AI/ML)</i> CGPA: 8.92	Vellore, Tamil Nadu 2023–Present
FIITJEE Junior College <i>Higher Secondary</i>	Hyderabad, Telangana 2021–2023
FIITJEE World School <i>Secondary</i>	Hyderabad, Telangana 2020

PROJECTS

Real-Time PPE Compliance Detection System <i>Python, PyTorch, OpenCV, torchvision, MobileNetV2, Scikit-learn</i> Built a real-time computer vision system from scratch to detect persons and verify PPE compliance (helmet, vest, gloves, boots, mask) in construction site and classify them as safe or not.
Smart Waste Segregation Using Machine Learning <i>TensorFlow, Keras, MobileNetV2, OpenCV, Deep Learning</i> Developed an image-based waste classification system using transfer learning with MobileNetV2, incorporating dataset merging, preprocessing, normalization, and data augmentation to improve robustness. The model was fine-tuned using TensorFlow/Keras with the Adam optimizer, achieving approximately 92–95% validation accuracy, and designed for real-time integration with camera-enabled or IoT-based smart bins.
StudyAI — AI-Powered PDF Study Assistant <i>Python, FastAPI, Groq LLM, JavaScript, ReactJS</i> Built a full-stack web app that converts PDFs into study notes, flashcards, MCQ quizzes, and mind maps using Llama via Groq API. Engineered a real-time chat interface for document QA, highlight-to-explain feature, and interactive quiz scoring with instant feedback
Structural Crack Monitoring System <i>Python, OpenCV, NumPy, Streamlit, Image Processing</i> Developed a computer vision-based structural crack monitoring system capable of crack detection, severity analysis, and temporal growth tracking. Implemented preprocessing, edge detection, morphological filtering, risk scoring, and an interactive dashboard for real-time visualization and analysis.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, HTML/CSS
Frameworks: TensorFlow, PyTorch
Developer Tools: Git, VS Code, GitHub, Figma, Wix Studio, Webflow
Libraries: pandas, NumPy, Matplotlib, Scikit-learn, OpenCV
Core CS: Data Structures and Algorithms, Database Management Systems, Object Oriented Programming

CERTIFICATIONS

Machine Learning Specialization <i>DeepLearning.ai</i>
Generative AI: Foundation Models and Platforms <i>IBM</i>
Generative AI Leader Professional Certificate <i>Google</i>